

```
#!/bin/bash
```

```
echo "QUESTION 1"
```

```
total=150
```

```
anonymous=$(echo "$total * 0.45" | bc)
```

```
smb=$(echo "$anonymous * 0.60" | bc)
```

```
spread=$(echo "$smb * 3 * 0.30" | bc)
```

```
total_compromised=$(echo "$anonymous + $spread" | bc)
```

```
risk=$(echo "$total_compromised / $total * 100" | bc -l)
```

```
echo "Initial systems compromised: $anonymous"
```

```
echo "SMB-susceptible systems: $smb"
```

```
echo "Expected second-stage spread: $spread"
```

```
echo "Estimated total compromised: $total_compromised"
```

```
echo "Percentage of network compromised: $risk%"
```

```
if (( $(echo "$risk > 70" | bc -l) )); then
```

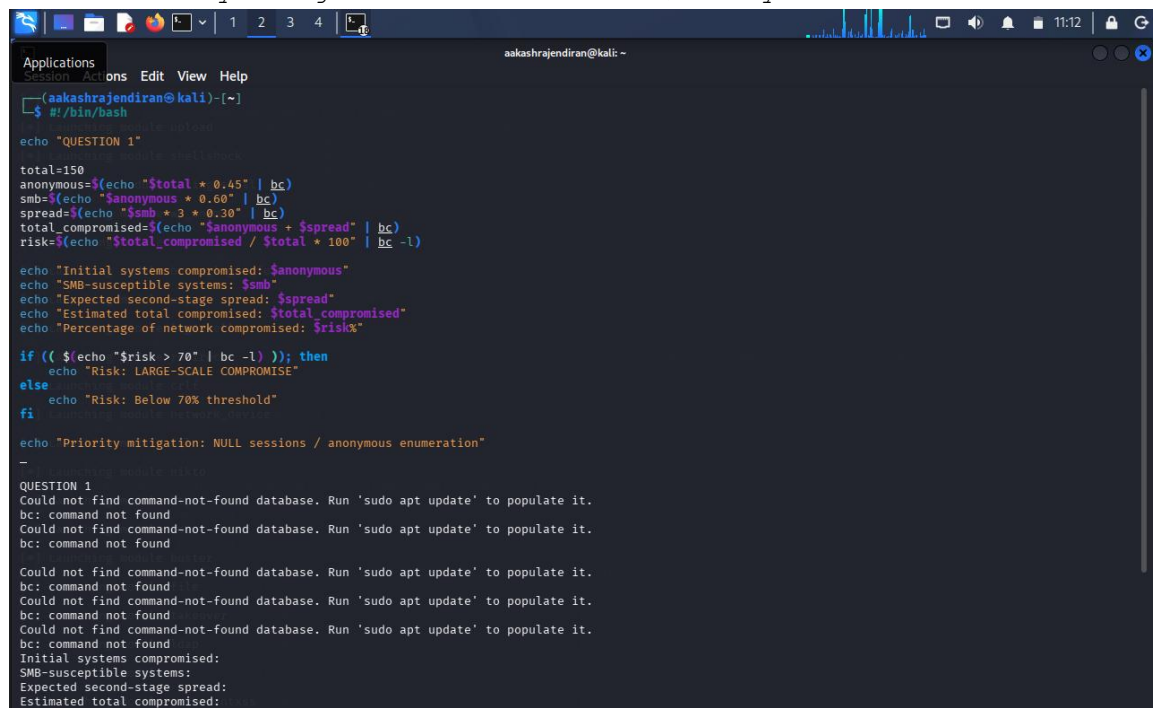
```
    echo "Risk: LARGE-SCALE COMPROMISE"
```

```
else
```

```
    echo "Risk: Below 70% threshold"
```

```
fi
```

```
echo "Priority mitigation: NULL sessions / anonymous enumeration"
```



```
Applications
aakashrajendiran@kali ~
aakashrajendiran@kali:~$ ./bin/bash
echo "QUESTION 1"

total=150
anonymous=$(echo "$total * 0.45" | bc)
smb=$(echo "$anonymous * 0.60" | bc)
spread=$(echo "$smb * 3 * 0.30" | bc)
total_compromised=$(echo "$anonymous + $spread" | bc)
risk=$(echo "$total_compromised / $total * 100" | bc -l)

echo "Initial systems compromised: $anonymous"
echo "SMB-susceptible systems: $smb"
echo "Expected second-stage spread: $spread"
echo "Estimated total compromised: $total_compromised"
echo "Percentage of network compromised: $risk%"

if (( $(echo "$risk > 70" | bc -l) )); then
    echo "Risk: LARGE-SCALE COMPROMISE"
else
    echo "Risk: Below 70% threshold"
fi

echo "Priority mitigation: NULL sessions / anonymous enumeration"

QUESTION 1
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found

Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
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bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Initial systems compromised:
SMB-susceptible systems:
Expected second-stage spread:
Estimated total compromised:
```

```
#!/bin/bash

echo "QUESTION 2"

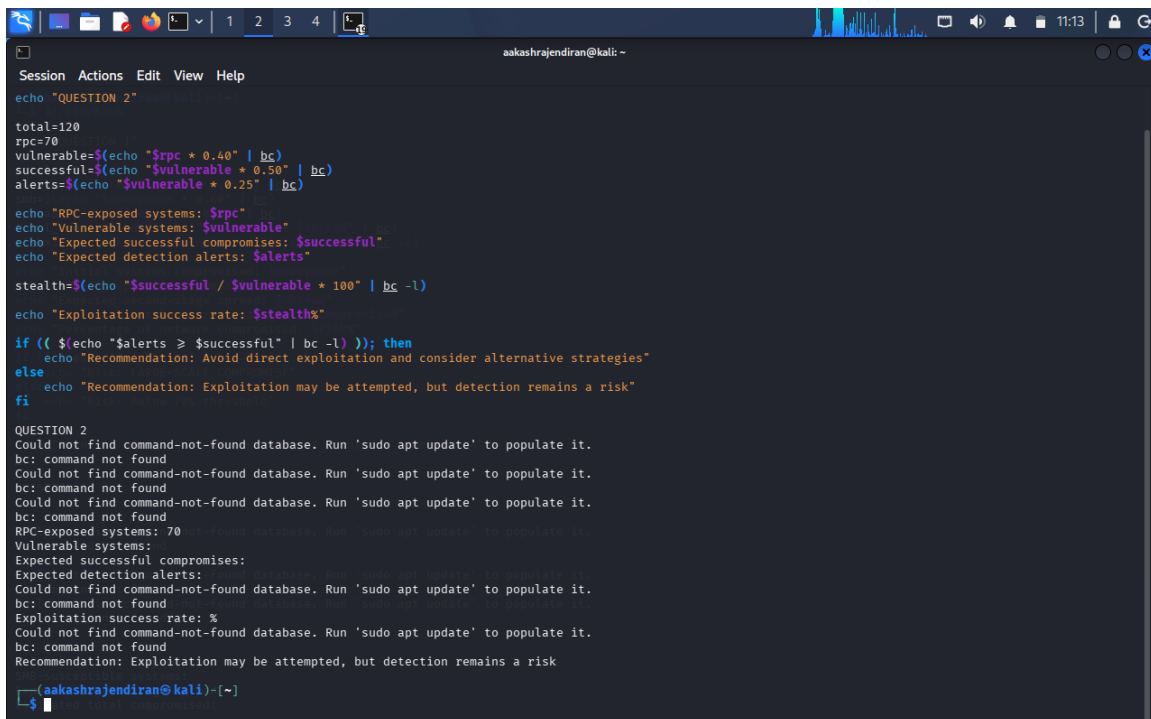
total=120
rpc=70
vulnerable=$(echo "$rpc * 0.40" | bc)
successful=$(echo "$vulnerable * 0.50" | bc)
alerts=$(echo "$vulnerable * 0.25" | bc)

echo "RPC-exposed systems: $rpc"
echo "Vulnerable systems: $vulnerable"
echo "Expected successful compromises: $successful"
echo "Expected detection alerts: $alerts"

stealth=$(echo "$successful / $vulnerable * 100" | bc -l)

echo "Exploitation success rate: $stealth%"

if (( $(echo "$alerts >= $successful" | bc -l) )); then
    echo "Recommendation: Avoid direct exploitation and consider
alternative strategies"
else
    echo "Recommendation: Exploitation may be attempted, but detection
remains a risk"
fi
```



```
aakashrajendiran@kali: ~
Session Actions Edit View Help
echo "QUESTION 2"

total=120
rpc=70
vulnerable=$(echo "$rpc * 0.40" | bc)
successful=$(echo "$vulnerable * 0.50" | bc)
alerts=$(echo "$vulnerable * 0.25" | bc)

echo "RPC-exposed systems: $rpc"
echo "Vulnerable systems: $vulnerable"
echo "Expected successful compromises: $successful"
echo "Expected detection alerts: $alerts"

stealth=$(echo "$successful / $vulnerable * 100" | bc -l)

echo "Exploitation success rate: $stealth%"

if (( $(echo "$alerts >= $successful" | bc -l) )); then
    echo "Recommendation: Avoid direct exploitation and consider alternative strategies"
else
    echo "Recommendation: Exploitation may be attempted, but detection remains a risk"
fi

QUESTION 2
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
RPC-exposed systems: 70
Vulnerable systems:
Expected successful compromises:
Expected detection alerts:
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Exploitation success rate: %
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Recommendation: Exploitation may be attempted, but detection remains a risk

aakashrajendiran@kali:~$
```

```
#!/bin/bash
```

```
echo "QUESTION 3"
```

```
p1=0.30
```

```
p2=0.20
```

```
p3=0.25
```

```
p4=0.15
```

```
permission=0.50
```

```
none=$(echo "(1-$p1)*(1-$p2)*(1-$p3)*(1-$p4)" | bc -l)
```

```
at_least_one=$(echo "1-$none" | bc -l)
```

```
full_compromise=$(echo "$at_least_one*$permission" | bc -l)
```

```
echo "Probability no service is vulnerable: $none"
```

```
echo "Probability at least one service is vulnerable: $at_least_one"
```

```
echo "Probability of at least one exploitable service: $at_least_one"
```

```
echo "Probability of full compromise: $full_compromise"
```

```
echo "At least one vulnerable service: $(echo "$at_least_one*100" | bc -l)%"
```

```
echo "Full compromise probability: $(echo "$full_compromise*100" | bc -l)%"
```

```
echo "At least one vulnerable service probability is calculated."
```

```
echo "File-permission hardening reduces privilege-escalation risk."
```

```
Session Actions Edit View Help
p4=0.15
permission=0.50

none=$(echo "(1-$p1)*(1-$p2)*(1-$p3)*(1-$p4)" | bc -l)
at_least_one=$(echo "1-$none" | bc -l)
full_compromise=$(echo "$at_least_one*$permission" | bc -l)

echo "Probability no service is vulnerable: $none"
echo "Probability at least one service is vulnerable: $at_least_one"
echo "Probability of at least one exploitable service: $at_least_one"
echo "Probability of full compromise: $full_compromise"

echo "At least one vulnerable service: $(echo "$at_least_one*100" | bc -l)%"
echo "Full compromise probability: $(echo "$full_compromise*100" | bc -l)%"

echo "At least one vulnerable service probability is calculated."
echo "File-permission hardening reduces privilege-escalation risk."
-

QUESTION 3
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Probability no service is vulnerable:
Probability at least one service is vulnerable:
Probability of at least one exploitable service:
Probability of full compromise:
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
At least one vulnerable service: %
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Full compromise probability: %
At least one vulnerable service probability is calculated.
File-permission hardening reduces privilege-escalation risk.
Could not find command-not-found database. Run 'sudo apt update' to populate it.
: command not found

(aakashrajendiran@kali)~$
```

```
#!/bin/bash
```

```
echo "QUESTION 4"
```

```
subnet1=0.40
```

```
subnet2=0.30
```

```
subnet3=0.20
```

```
root=0.70
```

```
none=$(echo "(1-$subnet1)*(1-$subnet2)*(1-$subnet3)" | bc -l)
```

```
unauthorized=$(echo "1-$none" | bc -l)
```

```
complete=$(echo "$unauthorized*$root" | bc -l)
```

```
echo "Probability no subnet gains access: $none"
```

```
echo "Probability at least one subnet gains access: $unauthorized"
```

```
echo "Probability of complete compromise: $complete"
```

```
echo "Unauthorized access probability: $(echo "$unauthorized*100" | bc -l)%"
```

```
echo "Complete compromise probability: $(echo "$complete*100" | bc -l)%"
```

```
echo "Recommendation: Restrict NFS access to only the required subnet."
```

```
echo "This reduces the number of possible unauthorized access paths."
```

```
Session Actions Edit View Help
aakashrajendiran@kali: ~
echo "QUESTION 4"

subnet1=0.40
subnet2=0.30
subnet3=0.20
root=0.70

none=$(echo "(1-$subnet1)*(1-$subnet2)*(1-$subnet3)" | bc -l)
unauthorized=$(echo "1-$none" | bc -l)
complete=$(echo "$unauthorized*$root" | bc -l)

echo "Probability no subnet gains access: $none"
echo "Probability at least one subnet gains access: $unauthorized"
echo "Probability of complete compromise: $complete"

echo "Unauthorized access probability: $(echo "$unauthorized*100" | bc -l)%"
echo "Complete compromise probability: $(echo "$complete*100" | bc -l)%"

echo "Recommendation: Restrict NFS access to only the required subnet."
echo "This reduces the number of possible unauthorized access paths."

QUESTION 4
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Probability no subnet gains access:
Probability at least one subnet gains access:
Probability of complete compromise:
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Unauthorized access probability: %
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Complete compromise probability: %
Recommendation: Restrict NFS access to only the required subnet.
This reduces the number of possible unauthorized access paths.

(aakashrajendiran@kali)~$
```

```
#!/bin/bash
```

```
echo "QUESTION 5"
```

```
records=50000
```

```
sensitive_percent=0.30
```

```
rate=500
```

```
block_time=40
```

```
extracted=$(echo "$rate * $block_time" | bc)
```

```
percentage=$(echo "$extracted / $records * 100" | bc -l)
```

```
sensitive=$(echo "$extracted * $sensitive_percent" | bc)
```

```
echo "Total database records: $records"
```

```
echo "Extraction rate: $rate records/second"
```

```
echo "Attack duration: $block_time seconds"
```

```
echo "Records extracted: $extracted"
```

```
echo "Percentage of database compromised: $percentage%"
```

```
echo "Estimated sensitive records exposed: $sensitive"
```

```
if (( $(echo "$percentage >= 30" | bc -l) )); then
```

```
    echo "Severity: HIGH"
```

```
else
```

```
    echo "Severity: MODERATE"
```

```
fi
```

```
echo "Detection helps limit the attack."
```

```
echo "SQL injection should be prevented through secure input handling."
```

```
Session Actions Edit View Help
block_time=40

extracted=$(echo "$rate * $block_time" | bc)
percentage=$(echo "$extracted / $records * 100" | bc -l)
sensitive=$(echo "$extracted * $sensitive_percent" | bc)

echo "Total database records: $records"
echo "Extraction rate: $rate records/second"
echo "Attack duration: $block_time seconds"
echo "Records extracted: $extracted"
echo "Percentage of database compromised: $percentage%"
echo "Estimated sensitive records exposed: $sensitive"

if (( $(echo "$percentage >= 30" | bc -l) )); then
    echo "Severity: HIGH"
else
    echo "Severity: MODERATE"
fi

echo "Detection helps limit the attack."
echo "SQL injection should be prevented through secure input handling."

QUESTION 5
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Total database records: 50000
Extraction rate: 500 records/second
Attack duration: 40 seconds
Records extracted:
Percentage of database compromised: %
Estimated sensitive records exposed:
Could not find command-not-found database. Run 'sudo apt update' to populate it.
bc: command not found
Severity: MODERATE
Detection helps limit the attack.
SQL injection should be prevented through secure input handling.

(aakashrajendiran@kali)-[~]
$
```